RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Tuesday, 25 August 2026, 8:52 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 8:59 AM
Time taken	6 mins 54 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    int count = 0;

    if (scanf("%d", &n) != 1) return 0;

    int i = 1; count++;
    int s = 1; count++;

    while (s <= n) {
        count++; // For the successful while condition evaluation

        i++;     count++;
        s += i;  count++;
    }
    count++; // For the final false while condition evaluation

    printf("%d\n", count);

    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Tuesday, 25 August 2026, 9:05 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 9:07 AM
Time taken	2 mins 27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    int count = 0;

    if (scanf("%d", &n) != 1) return 0;

    count++; // if(n == 1)
    if (n == 1) {
        count++; // printf("*")
    } else {
        for (int i = 1; count++, i <= n; i++) {
            for (int j = 1; count++, j <= n; j++) {
                count++; // printf("*") 1
                count++; // printf("*") 2
                count++; // break
                break;
            }
        }
    }

    printf("%d\n", count);
    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔

	Input	Expected	Got	
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Tuesday, 25 August 2026, 9:08 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 9:08 AM
Time taken	38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int num;
    int count = 0;

    if (scanf("%d", &num) != 1) return 0;

    for (int i = 1; count++, i <= num; ++i) {
        count++; // For if (num % i == 0) check
        if (num % i == 0) {
            count++; // For printf("%d ", i)
        }
    }

    printf("%d\n", count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Tuesday, 25 August 2026, 9:09 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 9:12 AM
Time taken	3 mins 13 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    int count = 1;

    if (scanf("%d", &n) != 1) return 0;

    for (int i = n / 2; count++, i < n; i++) {
        for (int j = 1; count++, j < n; j = 2 * j) {
            for (int k = 1; count++, k < n; k = k * 2) {
                count++;
            }
        }
    }

    printf("%d\n", count);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Tuesday, 25 August 2026, 9:14 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 9:14 AM
Time taken	12 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int count = 0;
    int rev = 0;
    int remainder;

    count++;
    count++;
    while (n != 0) {
        count++;
        remainder = n % 10;
        count++;
        rev = rev * 10 + remainder;
        count++;
        n /= 10;
        count++;
    }

    count++;
    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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